

How to Carry Out Exploratory Data Analysis with MySQL

Exploratory data analysis (EDA) reveals the internal pattern of data. It is often used to validate hypotheses, verify certain theories and explore data sets in order to find the distribution pattern of the sample collected. Summary statistics and visualisation of data are two frequently used data analysis parameters, which often help businesses to decide future courses of action.

In general, data is available in different forms and most of it is unstructured. For efficient and stable data processing, a primary requirement is to organise data into a well-defined form. Relational database management systems (RDBMSs) play this role and currently, MySQL is one of the most used among them. It provides a well-defined structured storage platform for data and forms the basis of exploratory data analysis or EDA. It also supports the conversion of different forms of data into its own form and vice versa. Data collected from other platforms may need to be refined and cleaned, and MySQL's pattern

recognition, filtering and conversion tools are helpful in this respect. Moreover, its strong interface with other programming language platforms like R, Python and PHP makes it a robust and most efficient EDA platform.

EDA with MySQL takes place over the following four steps:

1. The collection of data from different platforms
2. The cleaning and validation of data
3. Analysis of data
4. The export of data to other platforms for further analysis

Importing files

Since records of an organisation are always stored in different forms — starting from handwritten ledgers to WhatsApp communications, it's a Herculean task to convert data in these disparate formats into a structured form for an integrated higher-level analysis. MySQL supports conversion of data from Microsoft Excel, CSV, XML and pure text data to its own structured format. It also supports exporting of

Analysis of data
export DATA
cleaning

Analysis of data
export DATA
cleaning

